Dr. rer. nat. · Data Scientist at VALUNY

Jannik *Castenow*

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PROFILE

I build AI for statutory health insurance in Germany to improve care and to make communication with insured people faster and more personal. I hold a Ph.D. in theoretical computer science and have published 13 peer-reviewed papers on distributed algorithms.

EXPERIENCE

09/2023 – now

Data Scientist

VALUNY GmbH (formerly ITSC GmbH)

- Building AI that helps improve care for insured people
- Building AI that makes customer communication faster and more personal
- Developing and running ML pipelines, from deep learning and federated learning to gradient-boosted tree models
- Designing RAG architectures that bring LLMs into domain-specific workflows

01/2018 – 07/2023

Research Assistant

Heinz Nixdorf Institute, Paderborn University · Chair of Algorithms & Complexity

- Research on distributed algorithms and mobile robot coordination
- 13 peer-reviewed papers, talks at 9 international conferences
- Best Student Paper Award (SIROCCO 2020), Best Paper Awards (ICDCN 2020, SSS 2020, ALGOSENSORS 2021)
- Supervised student theses and tutorial groups

03/2017 – 12/2017

Student Research Assistant

Paderborn University

- Built a simulator for hybrid communication networks in C++
- Research on competitive routing in hybrid networks

10/2013 – 02/2017

Student Teaching Assistant

Paderborn University

Tutorials in Modeling, Data Structures and Algorithms, and Fundamentals of Programming 1

EDUCATION

2018 – 2023

Ph.D. (Dr. rer. nat.) in Theoretical Computer Science

Paderborn University · magna cum laude (1.0)

Dissertation: *Local Protocols for Contracting and Expanding Robot Formation Problems*

10/2015 – 10/2017

M.Sc. Computer Science

Paderborn University · Grade 1.0

07/2016 – 12/2016

Semester Abroad

University of Oklahoma, USA

10/2012 – 10/2015

B.Sc. Computer Science

Paderborn University · Grade 1.7

SKILLS

ML & Deep Learning	PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Hugging Face Transformers, LangChain, SHAP
Data Engineering	Python, PySpark, Polars, Narwhals, Pydantic
MLOps	Kubeflow, Kubeflow Pipelines, Docker, Kubernetes, MLflow
Visualization	Matplotlib, Seaborn, Plotly, Streamlit

CERTIFICATES

DeepLearning.AI	Deep Learning Specialization
Scrum.org	Professional Scrum Product Owner I, Professional Scrum Master I

LANGUAGES

German	Native
English	Fluent